User guide

FORGE User Guide

The seeded motif generator on SEQ TOOLS F5



FORGE ONE · Felucca 1.6.1 + FORGE 0.7 · M-VAVE FM-1
github.com/kkasemsa/FORGE-ONE (forge branch) · 22 October 2026
Felucca by Hügelton Instruments and contributors · GPL-3.0-only

FORGE 0.7 in FORGE ONE (Felucca 1.6.1) · M-VAVE FM-1

FORGE writes a musical idea onto a track in one key press, then lets that idea evolve: a melody on a synth track, a beat on a DRUM track (section 6). From 0.6, **STARTER** on the G5 key writes all four tracks at once, sounds included (section 7). FORGE lives in Felucca's SEQ TOOLS layer on the **F5** key. Every pattern comes from a **seed**: the same seed and settings always give the same steps, so you can note a seed you like and get the pattern back later.

All values in this guide are starting points. Judge the result by ear and turn the knobs until it sits right.

1. Quick start

1. Choose a track with **ALGORITHM**. This quick start is for a synth track; a DRUM track makes a beat instead (section 6).
2. Set the track's **ROOT** and **SCALE** on the SCL page, and its length with SEQ TOOLS' **K1 LEN**. FORGE reads all three, so set them first.
3. Press **PLAY** if you want to hear the result as you work.
4. Hold **SEQ** and tap **F5**. A pattern is written at once, and the screen shows it as a grid of steps.
5. Turn **K1 SEED** until you hear an idea you like. Then shape it with **K2 DENSITY**, **K3 EVOLVE** and **K4 STYLE**. Like the rhythm but not the notes? Press **C4** for new notes. Like the notes but not the rhythm? Press **D4**.
6. Press **OCT+** to keep it, or **OCT−** to put the old steps back.

If you change your mind after keeping a pattern, hold **SAVE** to undo the whole FORGE in one step, as long as you have not used another SEQ TOOLS key since.

2. Controls while FORGE is open

Control	What it does	Range
K1 SEED	Picks the pattern. Each seed is a different idea; turning past 99 wraps to 0	0–99
K2 DENSITY	How many of the style's steps sound; above 75 % it also adds notes off the style's rhythm	10–100 %, steps of 5
K3 EVOLVE	How much the idea changes across the pattern and from loop to loop (section 4)	0–4
K4 STYLE	The rhythm and pitch profile (section 3). Choosing a style sets DENSITY to that style's default; turning past AMBIENT wraps to BERLIN	8 styles
C4 (RANDOM)	New notes on the same rhythm. Each press is the next variation	NOTES 0–99
D4 (COOK)	A new rhythm; each step that still sounds keeps its pitch (with EVOLVE, later copies may drift differently)	RHYTHM 0–99
OCT+	Keep the pattern and close FORGE	
OCT−	Cancel: the track's steps go back to how they were before FORGE opened	

Control	What it does	Range
F5 again	Keep the pattern (same as OCT+)	
Another SEQ TOOLS key (not C4 or D4)	Keeps the FORGE pattern, then runs that tool	
Letting go of **SEQ**	Keeps the pattern	
SAVE held (afterwards)	Undo the whole FORGE: the steps and their locks come back (if no other tool ran since)	

Turning the knobs while the sequencer plays is safe: each step is replaced whole, so you never hear half-written steps. Each knob turn rewrites the pattern from the steps as they were before FORGE opened, so turning a knob back returns the earlier pattern exactly. The exception is STYLE: any turn of K4 sets DENSITY to the style's default, so set DENSITY after choosing the style.

What the screen shows. Each time FORGE opens or a knob turns, the header briefly shows `FORGE <seed> <style>`, for example `FORGE 42 BERLIN`. The four knob cards show SEED, DENSITY, EVOLVE and STYLE. Below them, every step of the pattern is a cell, 16 to a row:

- a **bright cell** is a note, with a small mark at its pitch (higher mark, higher note)
- a **half-bright cell** is a tie, holding the note before it
- a **dark cell** is a rest

Under the cells, **C4 NOTES** and **D4 RHYTHM** show how many times you have rerolled each (lit once above 0). Turning K1 SEED or K4 STYLE sets both back to 0.

The footer shows **OCT+ APPLY** and **OCT− CANCEL**. With MENU > SYSTEM > HELP on, a second row shows K1 SEED, K3 EVOLVE and K4 STYLE.

F5 lights up while FORGE is open.

3. The eight styles

Style	Knob card	Motif	Rhythm	Pitch	Octave	Default DENSITY
BERLIN	BERLIN	8 steps	16th ostinato, accents on 1 and 9	Small steps around the root	C3	85 %
DETROIT	DETROIT	16	Syncopated stabs	4-note scale chords	C4	70 %
DUB	DUB	16	Off-beats, long ties	3-note scale chords, narrow range	C3	55 %
MELODIC	MELO	16	8ths	Rises then falls in an arc	C4	75 %
ACID	ACID	16	16ths with accents and slides	Root pedal, other notes nearby, octave jumps	C2	75 %
TRANCE	TRANCE	8	Rolling 16ths	Arpeggios of the scale's triad and octave	C4	90 %
SYNTH	SYNTH	8	8ths	Root, fifth and octave, a third now and then	C3	90 %

Style	Knob card	Motif	Rhythm	Pitch	Octave	Default DENSITY
AMBIENT	AMBI	16	Very sparse, long ties	Wide intervals, some 3-note chords	C4	50 %

- **Motif** is the length of the idea before it repeats. If LEN is shorter than the motif, the motif is LEN steps long.
- **Octave** is where the root sits: the root note is the track's ROOT in that octave (A minor on ACID starts on A2). Melodies move a few scale steps above and below the root (AMBIENT more than an octave). Octave jumps only go up.
- **Step 1 is always the root**, so every pattern starts at home.
- **The scale decides the notes.** Every note is in the track's ROOT and SCALE. Chords, and the TRANCE and SYNTH arpeggios, are built from real intervals (a third, fifth and seventh above the note) kept inside the scale, so they work in every scale. In CHR the chords are minor (root, minor third, fifth, minor seventh); in the pentatonic scales the "seventh" is the scale's sixth; in WHOLE the chords are augmented.
- **The knob card** shortens MELODIC and AMBIENT so they fit. The header shows the full names.

Picking a sound for each style. These are starting points:

Style	Try it on
BERLIN	ANALOG or PHASE with a short decay; a slow LFO on the filter
DETROIT	WHEEL, FM6 or ANALOG pads with a short envelope; a 1/8D delay
DUB	ANALOG or FM6 chords with HALL reverb and a dotted delay; ECHO THROW in the FX layer
MELODIC	FM6, PHASE or TRIO leads
ACID	ANALOG with resonance up, MONO or LEGATO voice mode so the slides glide
TRANCE	ANALOG or TRIO with the gate short; the SLICER for movement
SYNTH	ANALOG or LOFI bass
AMBIENT	GRAIN, PHYS or FM6 pads with long release and HALL reverb

Chords need a polyphonic track: set the voice mode to **POLY** on EDIT > VOICE (K1). On MONO, LEG or UNI only one note of each chord sounds.

4. EVOLVE: how the idea changes

FORGE writes the motif once, then copies it through the track's LEN. EVOLVE decides how much the copies differ and how much each loop differs from the last.

EVOLVE	What happens
0	A pure loop. Every copy is the motif, with no chance, conditions or ratchets
1	Later copies sometimes sit a scale step higher or lower, and a few notes move a step. A few ornament notes get a 75 % or 50 % chance. ACID and TRANCE ratchet a few notes

EVOLVE	What happens
2	More drift. Some ornament notes get a **condition** instead of a chance: 2:2, 3:4, 2:4 or 4:4
3	As 2, and later copies also drop a note or add one in a rest
4	The most change: drift, rhythm changes and ornaments on more notes

- **The first copy keeps the motif's notes and rhythm.** Drift and note changes only happen in later copies, and build up the further along they are. The first copy can still get chances, conditions and (ACID, TRANCE) ratchets.
- **If LEN equals the motif length** (a 16-step style on LEN 16), there is only one copy, so EVOLVE only adds chances, conditions and ratchets. Use a longer LEN, or an 8-step style, to hear the copies drift.
- **Conditions** make one pattern change over several loops: a 4:4 note plays only on every fourth loop, and 2:2 only on every second. Each track counts its own loops from PLAY and from the start of each SONG section.
- **Chance** notes play on some loops and not others, so the line breathes.
- You can see and edit the chances and conditions FORGE wrote on **SEQ > DETAIL** (K1).

For long-form playing, try a short motif on a long LEN: for example BERLIN (8 steps) on LEN 64 at EVOLVE 2. The ostinato stays recognisable while the copies and conditions keep it moving.

5. Recipes

These are starting points. Each one lists the track settings to make first, then the FORGE settings.

Idea	Track settings	FORGE settings
Berlin School sequencer	ANALOG, LEN 32, DIV 1/16, A MIN	BERLIN, DENSITY 85, EVOLVE 2
Dub techno chord stab	FM6 or ANALOG, POLY, LEN 16, D MIN, HALL reverb	DUB, DENSITY 55, EVOLVE 1
Rolling acid line	ANALOG, MONO, LEN 16, resonance up, E MIN	ACID, DENSITY 75, EVOLVE 1, then turn SEED
Melodic techno lead	FM6 or PHASE, LEN 32, F# MIN	MELODIC, DENSITY 70, EVOLVE 2
Trance arpeggio	ANALOG or TRIO, LEN 16, gate short	TRANCE, DENSITY 90, EVOLVE 0
Ambient drift	GRAIN or PHYS, POLY, LEN 64, DIV 1/8, C MAJ	AMBIENT, DENSITY 50, EVOLVE 3

Working with a pattern you like. Note its seed, style and the C4 / D4 numbers. After you apply it, the steps are ordinary Felucca steps: edit single notes on SEQ > STEP, add parameter locks, or use the other SEQ TOOLS (REVERSE, SHIFT, COOK) on top.

6. FORGE on a drum track

From 0.5, F5 works on a DRUM track too. The same four knobs and the same eight style names make a **beat** instead of a melody. The map shows FORGE as a small **F** cell at the right end of the lanes row.

Control	On a DRUM track
K1 SEED	Picks the beat
K2 DENSITY	How many hits. Each style's main hits stay from 50 % up; above 75 % extra 16th hats and rim ghosts come in
K3 EVOLVE	0 is a pure loop. From 1, later bars move a hat or two and add ghost snares and rims with a chance; from 2 the last bar ends in a **snare roll fill on a 4:4 condition** (every 4th loop); from 3 kicks off the beat come and go
K4 STYLE	The beat (table below). Sets DENSITY to the beat's default
C4 (LOW)	New kick, snare, clap and tom; the hats stay
D4 (HATS)	New hats, rim and bell; the low lanes stay

The screen shows every step as a column of eight strips, KICK at the bottom and BELL at the top; a brighter strip is an accent. At LEN 33 to 64 the strips would be too thin, so each step shows two columns of marks instead: kick, snare, clap and tom on the left, hats, rim and bell on the right. The kick on step 1 is always there, and kicks on 1 and 9 are accented; snares and claps on 2 and 4 are accented half the time. Velocity follows the beat as on melodic tracks. Lane mutes (EDIT layer) and lane levels are not touched.

Style	Kick	Snare / clap	Hats	Extras	Default DENSITY
BERLIN	Four on the floor	None	Straight 8ths, 16ths at high density	Rim ticks, an open hat	70 %
DETROIT	Four on the floor, a push on 11	Clap on 2 and 4	8ths, open hat off the beat	Tom, rim	70 %
DUB	1 and 9, a push on 11	Rim on 2 and 4	8ths	Open hat	55 %
MELODIC	Four on the floor	Clap on 2 and 4	Off-beat 8ths, more at high density	Open hat, a tom on 16	70 %
ACID	Four on the floor, a push on 12	Snare on 2 and 4, clap	Off-beat 16ths	Open hat, rim; the most ratchets	75 %
TRANCE	Four on the floor	Clap on 2 and 4	8ths with open hats off the beat	Rim	85 %
SYNTH	1 and 7, a push on 12	Snare on 2 and 4	8ths	Open hat, tom	70 %
AMBIENT	1 only	None	Sparse	Open hat, tom, rim, a bell	45 %

Pick the kit first (PRESETS, or EDIT > K1 KIT): the same beat sounds very different on 80, DEEP or PERC. The beat is ordinary drum-grid steps once applied, so edit it on the grid, mute lanes in the EDIT layer, or run the lane tools over it.

7. STARTER: a song start on all four tracks

STARTER (SEQ TOOLS **G5**, from 0.6) is FORGE for the whole song. One press loads a sound on each of the four tracks and writes a pattern for each, all from one seed:

Track	Part
T1	BASS
T2	LEAD
T3	PAD (chords)
T4	DRUM (T4 becomes a DRUM track)

1. Select a track and set its **ROOT** on the SCL page. All four tracks take that ROOT. Leave SCALE on CHR to use the starter's own scale, or set one to use yours.
2. Hold **SEQ** and tap **G5** (on any track). Four sounds load and four patterns are written at once; the tempo goes to the starter's BPM (not when the FM-1 follows an external MIDI clock).
3. Turn **K1 SEED** for another song idea and **K4 STARTER** for another genre: each starter loads its own four sounds and tempo. **K2 DENSITY** makes all four busier or sparser (50–150 % of each part's own), and **K3 EVOLVE** works as in FORGE. **C4** gives all four new notes, **D4** new rhythms.
4. Press **OCT+** (or G5 again) to keep it, or **OCT−** to put all four tracks back exactly as they were: sounds, settings, steps, automation and tempo.

The screen shows a row a track: its preset and its first 16 steps. Under them, the C4 and D4 counts.

Starter	T1 BASS	T2 LEAD	T3 PAD	T4 DRUM	BPM	LEN (T1–T4)	SCALE*
BERLIN	ANALOG SQR BASS: 16th pedal sequence, octave jumps	TRIO ARP LEAD: a high, sparse line	GRAIN CLOUD PAD: one long chord a bar	DEEP KIT	120	16 16 32 16	MIN
DETROIT	FM6 FM BASS: off-beat root and fifth	ANALOG PWM STR: 4-note chord stabs	FM6 TINE EP: 4-note chords	TIGHT KIT	126	16 16 32 16	DOR
DUB	ANALOG SUB BASS: sparse, held	PHASE STRING: dub chord stabs	GRAIN SHIMMER: one chord a bar	DEEP KIT	118	16 16 32 16	DOR
MELODIC	TRIO FAT BASS: rolling off-beat 16ths	ANALOG SAW LEAD: a rising and falling line	FM6 PAD: long chords	DRUM KIT	122	16 32 32 16	MIN
ACID	ANALOG ACID: the 303 line, slides	LOFI ARP 8BIT: an arpeggio	ANALOG SOFT PAD: one chord a bar	TIGHT KIT	128	16 16 32 16	MIN
TRANCE	ANALOG SUB BASS: off-beat	TRIO ARP LEAD: the rolling triad arpeggio	ANALOG STRINGS: long chords	DRUM KIT	138	16 16 32 16	MIN
SYNTH	TRIO FAT BASS: 8th root and fifth	ANALOG BRASS: root-fifth riff	FM6 PAD: long chords	DRUM KIT	110	16 16 32 16	MIN

Starter	T1 BASS	T2 LEAD	T3 PAD	T4 DRUM	BPM	LEN (T1–T4)	SCALE*
AMBIENT	ANALOG SUB BASS: a drone, one note a bar	PHYS KALIMBA: wide, sparse notes	GRAIN CLOUD PAD: very long chords	PERC KIT	80	32 32 64 16	LYD

* The starter's SCALE is used only when the selected track's SCALE is CHR (the default). Any other SCALE you set is kept.

What STARTER changes: each track's sound (engine and factory preset, with its effect sends), its pattern, LEN, DIV (1/16), ROOT and SCALE, and the tempo. **It leaves alone:** each track's LEVEL, PAN and MUTE, the master and the effects' own settings. Steps that held something before and are now empty lose their parameter locks when you apply, as with FORGE; a sound load also drops automation of the old engine's own controls.

Undo: STARTER changes four tracks, and SAVE-held undo holds only one, so after STARTER it has nothing to undo. **OCT−** while STARTER is open is the way back. Save your project first if it matters.

Seeds: the same SEED, C4 and D4 numbers, STARTER, DENSITY, EVOLVE and ROOT (and SCALE, if you set one) always give the same four patterns. A set of starter seeds is pinned by a test, as FORGE's are.

CPU: each starter's four engines were chosen to play together, but FORGE ONE has not yet been tested on a real FM-1. If you hear voices drop out, swap the pad for a lighter engine (ANALOG or FM6) on the EDIT layer.

8. What FORGE writes, and what it leaves alone

Writes: steps 1 to LEN of the selected track. On a synth track each step gets notes (1, 3 or 4), NOTE, TIE or REST, accent, slide, ratchet (×2 or ×3), and a chance or condition. Velocity follows the beat: notes on the beat are louder and off-beat 16ths softer, with a little variation, between 40 and 124 (BERLIN, TRANCE and ACID the widest, DUB and AMBIENT the gentlest). Accented notes play at 127 as always. Every copy of the motif has the same velocities.

Leaves alone:

- steps past LEN
- the track's LEN, DIV, SWING and GATE, its sound, and its automation
- the other three tracks

Parameter locks: when you apply a pattern, steps that held something before and are now empty lose their locks, as with EUCLID. Undo (SAVE held) brings them back.

Compatibility: FORGE only uses fields that Felucca 1.6 already stores, so projects made with FORGE open on stock Felucca 1.6 and 1.6.1. Projects with conditional steps do not open on Felucca 1.5.1 or older. FORGE at EVOLVE 2 and up can write conditions, so check SEQ > DETAIL or AUTOMATION before going back.

Seeds and settings: the same SEED, C4 and D4 numbers, STYLE, DENSITY, EVOLVE, LEN, ROOT and SCALE always give the same steps. From FORGE 0.3 on, a set of seeds is pinned by a test, so a later version only gives a seed a different pattern when its changelog says so. Patterns you want to keep for

certain belong in a project. Change any of them and the pattern changes. At EVOLVE 0, changing DENSITY keeps the pitch of the notes that stay; with EVOLVE 1 and up it can also change how the later copies drift.

9. When it does not do what you expect

What you see	Why, and what to do
F5 does nothing	A SONG is playing ("STOP TO EDIT"), or the firmware is older than 0.5 on a DRUM track
G5 does nothing	A SONG is playing ("STOP TO EDIT"), or the firmware is older than 0.6
STARTER changed my sounds and I want them back	Press **OCT−** before letting go of SEQ. After applying, there is no undo for STARTER: load your project again
The notes are too low or too high	Each style has its own octave. To move a pattern, set **QNT** to **SEQ+T** on the SCL page and turn **TRN** (TRN alone, and the octave buttons, only move the keys). Or try a style in another octave
Chords play as single notes	The voice mode is MONO, LEG or UNI. Set **POLY** on EDIT > VOICE
ACID slides do not glide	Use MONO or LEG voice mode, and some GLIDE on EDIT > VOICE (K2)
The pattern changes every loop and I want it fixed	Set EVOLVE to 0, or remove chances and conditions on SEQ > DETAIL
The knobs edit LEN, not SEED	FORGE is not open (F5 is not lit). Tap F5 while holding SEQ
I applied it and want the old steps back	Hold **SAVE** before using another tool (one undo). Or press **OCT−** before letting go of SEQ: it puts back everything the layer changed since it opened, LEN, DIV, SWING and GATE included

10. Credits

FORGE is part of FORGE ONE, a fork of Felucca by Leo Kuroshita, Hügelton Instruments, and contributors. GPL-3.0-only. Source: github.com/kkasemsa/FORGE-ONE, `forge` branch; FORGE's code is in `firmware/src/ui_forge.c`, STARTER's in `firmware/src/ui_starter.c`.